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How Much of Genomic Language Model Variant-Effect Prediction Is Base Composition?

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Abstract

Zero-shot variant-effect prediction (VEP) with genomic language models (gLMs) scores a variant by the log-likelihood ratio LLR $= \log p(\text{alt}) - \log p(\text{ref})$. This quantity mixes learned regulatory function with a purely local signal, namely how mutationally expected a substitution is given its base composition and sequence context. We isolate and quantify this composition/mutation-expectedness axis as a confound on real clinical and functional VEP benchmarks, across eight gLMs and four benchmark families, using 19 hand-specified features that read local base composition and mutational context and nothing else. A gradient-boosted classifier on those features, fitted to labels under leave-one-chromosome-out cross-validation, reaches AUROC 0.720 on noncoding ClinVar and recovers 45–53% of the above-chance AUROC of every leading method, including Evo2-40B, GPN-MSA, CADD and phyloP; the result holds when coding-overlapping variants are removed. A label-free version of the same test, scoring only the part of a model's own likelihood ratio that composition predicts, still recovers 31–38%. On composition-matched benchmarks (TraitGym) recovery falls to roughly a fifth to a quarter; the two regimes differ in ascertainment as well as in matching, so we read the contrast as a bound on how much apparent performance is composition-attributable. The confound is strongly model dependent: smaller single-sequence gLMs (Nucleotide Transformer, Caduceus, HyenaDNA) do not exceed the composition baseline on any benchmark, while alignment-based (GPN-MSA) and large autoregressive (Evo2-40B) models retain signal that survives composition matching and is enriched for TF-motif disruption (pooled odds ratio 3.1, $p < 10^{-3}$), though that surviving signal is not separable from conservation. Within saturation-mutagenesis screens with fixed background the leakage is negligible, which localises the confound to across-set ascertainment rather than local context. We provide a model-agnostic post-hoc pentanucleotide calibration that halves the mutation-rate coupling of the most affected model at negligible AUROC cost and shrinks its CpG versus non-CpG false-positive-rate gap from 5.8 to 0.7 points, and release it as a drop-in evaluation transform.

Keywords: genomic language models, variant effect prediction, benchmark confounds, base composition, mutation rate, calibration

1. Introduction

Genomic language models (gLMs) are increasingly used as zero-shot predictors of noncoding variant effect: a pretrained model scores a single-nucleotide variant (SNV) by the log-likelihood ratio LLR $= \log p(\text{alt}) - \log p(\text{ref})$, ranked against clinical or functional labels (Benegas et al., 2025a; Brixi et al., 2026). Aggregate AUROCs on noncoding ClinVar routinely exceed 0.95, read as evidence that the models have learned regulatory grammar.

Recent work complicates that reading. On synthetic promoters, gLM compositional sensitivity tracks AT content almost perfectly and a small position-aware model beats billion-parameter gLMs (Anonymous, 2026); stratifying ClinVar by region type collapses Evo2's aggregate noncoding AUROC, because pathogenic variants concentrate where prediction is easiest, canonical splice sites above all (Lu et al., 2025); cyclic permutation of flanking context collapses Evo2 tRNA sensitivity (Mathur and Sachidanandam, 2026); frozen gLM representations give little advantage over one-hot encodings on regulatory tasks (Tang et al., 2025), with usable information concentrated in embedding rather than transformer layers (Rochkoulets et al., 2026).

These identify positional-grammar, region-type and context-artifact confounds. We isolate a distinct axis on real VEP: local *base composition* and *mutational expectedness*. A model's likelihood partly encodes how expected a substitution is under neutral evolution, since CpG transitions mutate an order of magnitude faster than transversions and mutation subtype couples to CpG and GC content (Carlson et al., 2018). Because pathogenic variants are ascertained non-randomly across mutation types, this component of LLR tracks labels through the ascertainment pathway rather than through function, which is why CADD and GPN-MSA prefer gnomAD-common over ClinVar-benign controls (Rentzsch et al., 2019; Benegas et al., 2025a). GPN-Star found raw gLM scores track pentanucleotide mutation rate at $\rho = 0.31$–0.34 and calibrated a single model at training time (Benegas et al., 2025c).

Contributions. We turn that observation into a benchmark-wide, cross-model, post-hoc audit. (1) A composition baseline recovers about half of the above-chance AUROC of every leading model on unmatched ClinVar and about a fifth on matched TraitGym, an upper bound on compositional overlap rather than a causal decomposition, bracketed from below by a label-free estimate of the same quantity (§4.2). (2) The confound is model-heterogeneous: single-sequence gLMs do not exceed the baseline, while alignment-based and large autoregressive models keep signal that survives composition matching, though little that survives a conservation baseline (§4.1, §4.3). (3) Within-element saturation mutagenesis localises the confound to across-set ascertainment rather than local context (§4.4). (4) We generalise pentanucleotide calibration into a model-agnostic post-hoc transform and release it (§4.5). (5) We map where the surviving signal sits (§4.6). The framing follows the confound-audit genre of Gupta et al. (2025): name a hidden confound, build a clean evaluation, measure its effect across a model class, and localise where real signal remains.

2. The composition and mutation-expectedness confound

We define two entangled axes for an SNV, both hand-specified summaries of the reference genome that involve no representation learning. *C1, local composition change*: for window half-width w, Δ_k is the L_1 distance between the k-mer frequency spectra of the $\pm w$ reference and alternate sequences ($k \in \{1, 2, 3\}$) and ΔGC their GC difference, swept over $w \in \{11, 21, 51\}$ bp; these read out to ± 51 bp but keep only composition, discarding all positional information. *C2, mutational expectedness*: the pentanucleotide neutral mutation rate for the substitution, a direct lookup of the reference pentamer and alternate base in the Carlson et al. ERV table with nothing fitted (Carlson et al., 2018); the CpG status of the reference and alternate sites; the transition/transversion class; and the flanking GC background. All 19 features are strand-invariant.

The quantity we audit is LLR $\approx \alpha\cdot$(function)$+\beta\cdot$(expectedness)$+\varepsilon$ (schematic in Supplementary Figure S1): the confound is the β term, which tracks labels through ascertainment rather than function, and matching and calibration target it. We do not assume the linear form holds. Where the split matters we estimate the expectedness term without labels, by regressing each model's own LLR on the composition features and scoring the fitted component and its residual separately (§4.2).

3. Experimental setup

Benchmarks. *ClinVar noncoding* (153,508 SNVs, 25,571 pathogenic): $\geq$2-star pathogenic/likely-pathogenic versus benign/likely-benign SNVs overlapping an ENCODE candidate cis-regulatory element (cCRE) (ENCODE Project Consortium, 2020); unmatched and region-heterogeneous, so the confound should be largest. *TraitGym* Mendelian (3,380; 338 positive) and complex (11,400; 1,140)

(Benegas et al., 2025b): fine-mapped causal regulatory variants with nine controls per positive matched on chromosome, consequence and distance to transcription start site (TSS), plus minor allele frequency (MAF) and linkage disequilibrium (LD) for complex traits, so the confound should be smallest. *Kircher* saturation massively-parallel reporter assay (MPRA) (Kircher et al., 2019): 39,170 SNVs across 21 elements with a continuous $\log_2$ effect and fixed within-element background. *GLRB causal expression QTL* (InstaDeep et al., 2024): 8,850 held-out SNVs where zero-shot DNA LMs are known to be near chance.

Models. We use authors' released per-variant scores throughout, so model quality is not confounded by our inference. TraitGym supplies eight gLMs (GPN-MSA, alignment-based; GPN, its single-sequence counterpart; Evo2-7B/40B; Nucleotide Transformer; Caduceus; HyenaDNA; SpeciesLM) plus CADD, phyloP and phastCons; on ClinVar we use precomputed Evo2-7B/40B scores, a genome-wide GPN-MSA tabix lookup, CADD, phyloP and SpliceAI, so the alignment-based anchor GPN-MSA appears on every benchmark. CADD is a supervised integrative score trained partly on ClinVar-adjacent labels, not a self-supervised gLM; we mark it (†) and read it as an integrative reference. Our composition baseline is a histogram gradient-boosted-tree classifier (250 trees, learning rate 0.05) on the 19 C1/C2 features, with no learned representation and no positional sequence model, but fitted to benchmark labels under the same leave-one-chromosome-out folds as everything else. It is therefore not a zero-shot score, and we report label-free comparators next to it in §4.2. A small one-hot CNN on ±100 bp windows serves as a learned-from-scratch reference.

Protocol. We report AUROC and area under the precision-recall curve (AUPRC) under leave-one-chromosome-out cross-validation with sample-size-weighted averaging (Benegas et al., 2025b), partial Spearman correlations controlling for consequence or region, and percentile bootstrap intervals; the composition classifier uses the same folds, so no variant informs its own fold. Each score carries a single fixed orientation per benchmark (the sign giving AUROC ≥ 0.5), chosen once at the benchmark level as in TraitGym and never per fold or per variant, and |LLR| enters only the E1 correlations, so no orientation freedom is used to inflate a model. The sign can almost always be prespecified instead from each score's own convention (deleterious means low LLR, high conservation): that rule agrees in 24 of 29 model×benchmark cells, and the five disagreements are a released file with an inverted convention (Evo2 on ClinVar, already higher-means-pathogenic) or models within 0.02 AUROC of chance, where the sign is noise. Significance uses the paired DeLong test with Benjamini–Hochberg control over the composition-versus-model comparisons in Table 1, one per model×benchmark cell.

4. Results

4.1. E1: gLM scores encode the confound, model-dependently

Figure 1(a) reports the partial Spearman correlation between |LLR| and the pentanucleotide mutation rate, controlling for consequence. GPN-MSA is most strongly coupled ($\rho = -0.33$ on ClinVar, -0.16 to -0.20 on TraitGym), consistent with an alignment model that tracks neutral rate; Evo2-40B is intermediate (-0.08 to -0.18), Evo2-7B weaker, and the smaller single-sequence models (Nucleotide Transformer, Caduceus, HyenaDNA) near zero. Conservation baselines couple similarly, sometimes with the opposite sign (phyloP-447way $+0.16$). E1 is descriptive of each score, saying which models encode the neutral spectrum rather than how much of a benchmark's label signal is compositional, which is E2's question.

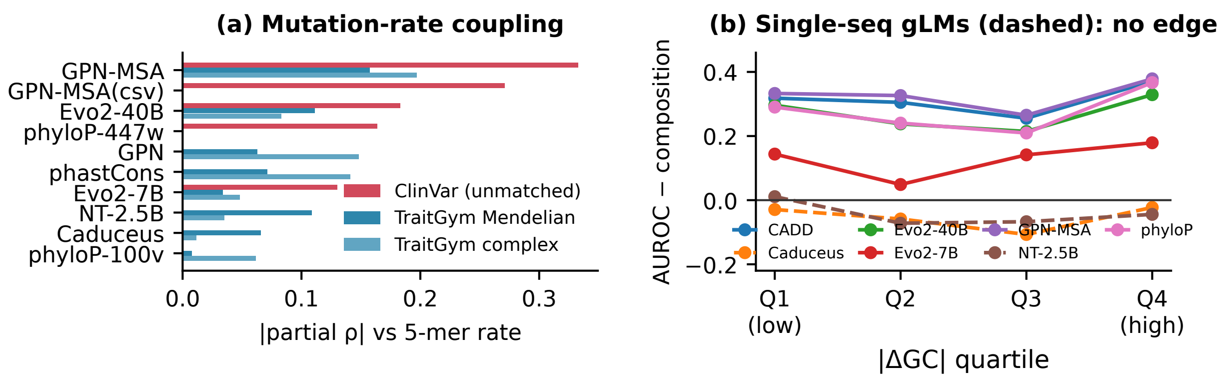

Figure 1: **Confound structure.** (a) Partial $|\rho|$ between |LLR| and the 5-mer mutation rate per model and benchmark. GPN-MSA is most coupled. (b) Collapse curve on TraitGym Mendelian: model AUROC minus composition AUROC within $|\Delta\text{GC}|$ quartiles. Smaller single-sequence gLMs (dashed) hold no edge over composition in any bin; alignment-based and large models keep a stable edge.

This is not an artifact of the labelled sets. At genome-wide neutral sites GPN-MSA's mean score per pentanucleotide context tracks the Carlson neutral rate at $\rho = 0.63$ (3,072 contexts, 3.6×10^7 scores; CpG contexts -0.10 versus -1.33 elsewhere): the model has learned the neutral spectrum, the very component that leaks into LLR. Nor is the high-rate tail a corner case: counting all 2.76×10^9 autosomal pentamers, the top decile of contexts by Carlson rate covers 19% of genome positions but holds 42% of ClinVar noncoding variants and 29–31% of TraitGym's. Ascertainment then makes the coupling leak into labels: ClinVar benign SNVs sit at higher mutation-rate contexts than pathogenic ones (0.033 versus 0.028, $p < 10^{-140}$) and are more often transitions (76% versus 65%) and CpG (35% versus 31%), so a score encoding the neutral spectrum separates the classes with no regulatory understanding.

4.2. E2: a composition baseline recovers much of aggregate performance

On ClinVar the composition classifier reaches AUROC 0.720 (per-chromosome bootstrap 95% confidence interval, CI, 0.716–0.723). Call a method *leading* when it beats that baseline by at least 0.05 AUROC at BH-adjusted DeLong $p < 0.05$: on ClinVar GPN-MSA (0.958), Evo2-7B (0.960), Evo2-40B (0.963), CADD (0.986) and phyloP (0.912), so the recovery ratio $(\text{comp} - 0.5)/(\text{model} - 0.5)$ is 0.45–0.53 for every one (all $p \approx 0$). On matched TraitGym the baseline reaches only 0.586 (Mendelian, CI 0.555–0.617) and 0.520 (complex, CI 0.502–0.538, only just clear of chance), with ratios of 0.18–0.25 for the leading methods; weaker gLMs score higher ratios because they sit closer to the baseline, and on GLRB causal eQTL the baseline (0.596) exceeds GPN-MSA (0.560). The composition intervals on ClinVar and on the matched sets do not overlap (Table 1, Figure 2). We read that gap as a bound on how much apparent performance is composition-attributable, not as a subtraction: the two regimes differ in ascertainment, positive definition and control sampling as well as in matching. The ratio's denominator is also the model's own above-chance AUROC, so a weaker method mechanically earns a higher ratio: phyloP's 0.53 on ClinVar exceeds GPN-MSA's 0.48 because phyloP scores 0.912 against 0.958, not because it is more composition-coupled, its coupling having the opposite sign (§4.1). E1 describes individual scores; E2 is a property of the benchmark's label structure.

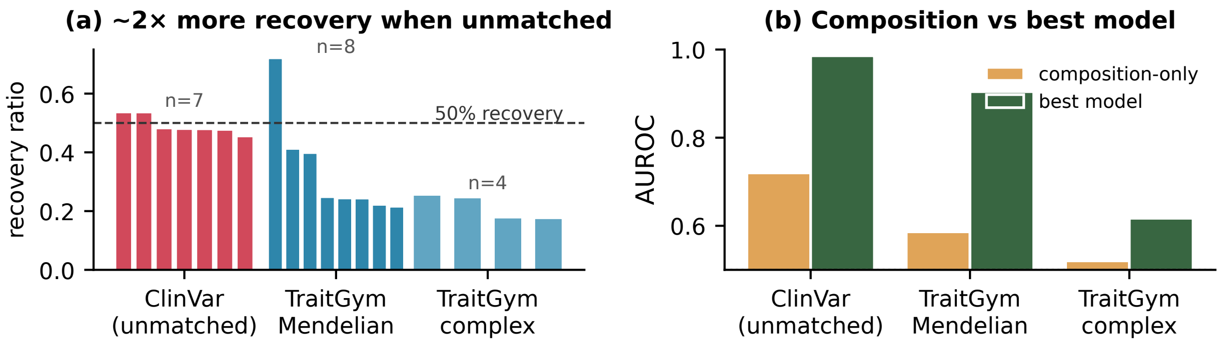

Figure 2: **Composition recovery.** (a) Recovery ratio per model, grouped by benchmark; composition recovers about twice as much above-chance AUROC on unmatched ClinVar as on matched TraitGym. (b) Absolute AUROC of the composition baseline versus the best model per benchmark.

The ClinVar effect is not an artifact of region composition (full tables in the supplement). Excluding the 46,852 coding-annotated variants leaves the baseline at 0.734 with ratios of 0.48–0.50, and removing the 9,067 splice-proximal variants *raises* it to 0.722 with every ratio slightly higher, so the confound is not carried by the easy splice variants that concern Lu et al. (2025). Bootstrapping the ratio itself gives GPN-MSA 0.48 (0.47–0.49) on ClinVar against 0.21 (0.14–0.29) on Mendelian, non-overlapping, and swapping benign controls for gnomAD common variants raises the baseline to 0.777.

Since the classifier is label-fitted, we also report label-free comparators. Single features with signs prespecified from mutation-rate theory are weak: $-\mu_5$ reaches 0.550 on ClinVar, 0.545 on Mendelian, 0.504 on complex traits. Much stronger is the composition-predictable part of a model's *own* score. Regressing GPN-MSA's ClinVar LLR on the 19 features leave-one-chromosome-out, with no label anywhere in the fit, explains $R^2 = 0.15$ of its variance, and that fitted component alone reaches AUROC 0.641, or 31% of GPN-MSA's above-chance AUROC recovered without a single label (Evo2-40B $R^2 = 0.07$, 0.676, 38%); the residual keeps 0.946. This estimates the α/β split of §2 rather than assuming it, and brackets the confound from both sides: 0.31–0.38 label-free against 0.45–0.53 label-fitted. Composition also explains far more of the single-sequence models' scores than of GPN-MSA's ($R^2 = 0.34$ for Nucleotide Transformer against 0.13 on complex traits), which is E3a's collapse seen from the score side.

Combining features under the same folds separates unique contributions. Adding GPN-MSA to composition raises ClinVar AUROC from 0.717 to 0.971 (+0.254), while adding composition to GPN-MSA adds +0.013 (Evo2-40B: +0.193 versus +0.057): the leading models subsume composition and carry large independent signal on top of it, whereas single-sequence gLMs do not exceed it. The confound inflates the apparent margin between a model and a trivial baseline, which is what a leaderboard column communicates, without making the leading models reducible to composition.

4.3. E3: matching shrinks but does not erase the gLM edge

Figure 1(b) bins TraitGym Mendelian variants by $|\Delta\mathrm{GC}|$ and plots each model's AUROC minus the composition AUROC. Nucleotide Transformer, Caduceus and HyenaDNA hold no edge in any bin, so their apparent VEP signal is composition; GPN-MSA, Evo2-40B, CADD and phyloP keep a stable edge of roughly 0.24.

Table 1: Leave-one-chromosome-out AUROC with per-chromosome stratified-bootstrap 95% CIs (300 resamples; 1,000 for the composition rows, which the text also quotes), AUPRC, and the composition recovery ratio $(\text{AUROC}_{\text{comp}} - 0.5)/(\text{AUROC}_{\text{model}} - 0.5)$ versus each model. A ratio above one (marked >1) means the model does not exceed composition. Composition and model AUROC CIs are non-overlapping on every matched-vs-unmatched contrast. DeLong tests are significant ($p < 0.05$; $p \approx 0$ for the leading models) except where a model sits at chance on complex traits (Evo2-40B, NT; $p > 0.2$). † CADD is a supervised integrative score (trained partly on ClinVar-adjacent labels), shown as an integrative reference rather than a zero-shot gLM. Evo2-7B and the remaining single-sequence gLMs are in Supplementary Table S1.

Method	AUROC [95% CI]	AUPRC	Recovery
ClinVar (unmatched)			
Composition-only	0.720 [0.716,0.723]	0.362	–
GPN-MSA	0.958 [0.957,0.960]	0.861	0.48
Evo2-40B	0.963 [0.962,0.965]	0.870	0.47
CADD†	0.986 [0.985,0.987]	0.958	0.45
phyloP	0.912 [0.909,0.914]	0.775	0.53
TraitGym Mendelian (matched)			
Composition-only	0.586 [0.555,0.617]	0.153	–
GPN-MSA	0.904 [0.886,0.921]	0.694	0.21
Evo2-40B	0.851 [0.830,0.872]	0.557	0.25
NT-2.5B	0.534 [0.508,0.566]	0.120	>1
Caduceus	0.509 [0.483,0.539]	0.108	>1
CADD†	0.893 [0.871,0.914]	0.712	0.22
phyloP	0.857 [0.830,0.883]	0.655	0.24
TraitGym complex (matched)			
Composition-only	0.520 [0.502,0.538]	0.115	–
GPN-MSA	0.580 [0.562,0.600]	0.212	0.25
Evo2-40B	0.516 [0.498,0.534]	0.137	1.26
NT-2.5B	0.516 [0.500,0.534]	0.114	>1
CADD†	0.616 [0.598,0.637]	0.250	0.18
phyloP	0.583 [0.565,0.604]	0.184	0.24
GLRB eQTL			
Composition-only	0.596 [0.585,0.607]	0.633	–
GPN-MSA	0.560 [0.548,0.571]	0.599	>1

We then add our own matching on top of what each benchmark already provides. TraitGym's inherited matching pairs each positive with nine controls on chromosome, consequence and TSS distance, plus MAF and LD for complex traits; ClinVar has none. Our added step matches each positive to one distinct control by nearest neighbour in standardised Euclidean distance over five composition features (μ_5, ΔGC_{51}, reference CpG, transition, flanking GC), searching within the same consequence or cCRE class and taking the nearest of five candidates that is not already used. No positive is dropped on TraitGym (338 and 1,140 positives retained, giving 1:1 subsets of 674 and 2,280); on ClinVar 24,353 of 25,571 positives find an unused control, for a subset of 49,924. On these subsets model AUROC drops only slightly (GPN-MSA −0.006 on ClinVar; Evo2-40B −0.019, NT −0.025 on Mendelian) and the ranking reorders mildly (Kendall $\tau = 0.81$–0.91). The models that clear chance therefore carry composition-independent signal.

Conservation is the harder baseline. GPN-MSA is trained on the Multiz alignments that phyloP and phastCons are computed from, so conservation, not composition, is the rival explanation for whatever survives matching. We therefore rerun the incremental test of §4.2 with conservation in place of composition (phyloP-100way and -447way on ClinVar; phyloP-100way, phyloP-241mammal and phastCons on TraitGym). Conservation is by far the stronger baseline: AUROC 0.924 on ClinVar, 0.843 on Mendelian and 0.599 on complex traits, against 0.710, 0.586 and 0.520 for composition. Measured against it the gLM increment shrinks by roughly a factor of five: adding GPN-MSA to conservation buys +0.039 on ClinVar and +0.042 on Mendelian, against +0.254 and +0.306 over composition, Evo2-40B buys +0.051 and +0.056, and on complex traits GPN-MSA adds nothing (−0.001). Conservation in turn adds almost nothing on top of GPN-MSA (+0.006, −0.001, +0.009), so the alignment-based model subsumes the conservation tracks and adds a small increment of its own. What survives composition matching is therefore mostly, though not entirely, conservation, and on complex traits it is entirely conservation.

4.4. E4: within fixed background the leakage is negligible

The Kircher screens fix the background within each element, isolating the substitution and its immediate context. For each of the 21 elements we compare the raw Spearman correlation between GPN-MSA LLR and the measured $\log_2$ effect with a partial correlation controlling for substitution type, ΔGC and CpG status. The two nearly coincide (mean absolute change below 0.02; Supplementary Figure S2(a)), so within a fixed background the model–assay correlation is not a composition artifact, which with E2 localises the confound to across-set ascertainment rather than local context.

4.5. E5: a model-agnostic pentanucleotide calibration

What the transform is. For a variant with reference pentamer c and alternate base a, let $\mathrm{ctx} = (c, a)$ index one of the 3,072 mutation contexts. The transform subtracts a neutral reference score for that context, on the score's own raw scale and with no rescaling: $s_{\mathrm{cal}} = s - \hat{s}_{\mathrm{neutral}}(\mathrm{ctx})$. Being a per-context shift it leaves the ranking *within* a context untouched, so all reordering is between contexts. The *control-based* estimator averages the score over control variants sharing ctx, leave-one-chromosome-out so no variant contributes to its own reference, with contexts absent from a fold falling back to their transition/transversion × CpG stratum (146 of 153,507 ClinVar variants, 0.1%). The *label-free* estimator, after Benegas et al. (2025c), averages over a genome-wide neutral sample (250 windows of 50 kb, 3.6×10^7 scored substitutions), covering all 3,072 contexts with no labels and no fallback.

What it does. On ClinVar the control-based transform halves GPN-MSA's mutation-rate coupling (partial ρ from −0.33 to −0.17) and reduces Evo2-40B's, at an AUROC cost below 0.005 (Supplementary Figure S3); the label-free variant cuts the coupling to 0.18 with AUROC unchanged (0.958 to 0.961), and the leaderboard reorders mildly (Kendall $\tau = 0.80$ on ClinVar, 0.91 on TraitGym). Leaving AUROC alone shows the transform is harmless, not useful, so we measure one thing it fixes: error *disparity* across contexts. Thresholding raw GPN-MSA at a global 10% false-positive rate, benign CpG variants are flagged at 6.2% and benign non-CpG variants at 11.9%, a 5.8-point gap; after calibration the gap is 0.7 points (10.5% versus 9.8%) and sensitivity at that threshold rises from 0.918 to 0.932, so a single ClinVar operating point means much more nearly the same thing in both context classes. Consistent with E4 it buys nothing where there is nothing to remove: on the fixed-background Kircher screens the rank correlation with measured effects is unchanged ($|\rho|$ 0.119 to 0.116, 12 of 21 elements improve, Wilcoxon $p = 0.95$), and on the five single-sequence gLMs, which carry almost no coupling, it shifts AUROC by at most 0.023 and leaves their ranking intact. Unlike GPN-Star (Benegas et al., 2025c), which calibrates one model at training time, ours is post-hoc and applies to any released score vector. We release it as a drop-in `calibrate_llr` function with the per-context neutral tables.

4.6. E6: the surviving signal, and what it shares with conservation

Among positives we compare variants the calibrated GPN-MSA ranks highly but the composition baseline misses ("gLM-wins") against the reverse ("composition-wins"). Concretely, on percentile ranks taken over all variants in the benchmark under the fixed orientation, a positive is a gLM-win when its gLM rank exceeds 0.7 and its composition rank falls below 0.5, and a composition-win under the mirrored condition. gLM-wins are enriched for overlap with transcription-factor (TF) motifs and ChIP footprints (Supplementary Figure S2(b)): pooled over both trait sets the odds ratio is 3.1 (Fisher 95% CI 1.8–5.3, label-permutation $p < 10^{-3}$), 2.3 on complex traits alone (1.3–4.0, $p = 0.005$), and 4.7 but underpowered on the smaller Mendelian set (0.8–46, $p = 0.28$, only 13 composition-wins). Sweeping both thresholds over seven settings from $(0.6, 0.5)$ to $(0.9, 0.5)$, the complex-trait odds ratio stays above one throughout (1.6–3.0) and is significant in five; the Mendelian estimate is above one in all seven but its control group falls to as few as three variants. The direction is robust to the threshold, the magnitude is not.

Conservation is again the competing explanation, and here it is not ruled out. Adjusting the enrichment for phyloP in a logistic model attenuates the complex-trait odds ratio from 2.31 to 1.87 (bootstrap CI 0.99–3.70), and substituting phyloP for composition in the win definition removes it (OR 1.05, CI 0.53–2.01 on complex; 1.67, 0.74–4.71 on Mendelian). The defensible claim is thus narrower than gLMs uniquely seeing motif disruption: what survives composition decontamination is motif-enriched *relative to composition* and is not separable from conservation here.

5. Discussion

Our claim is specific, not a blanket dismissal. A hand-specified composition baseline recovers about half of the apparent zero-shot VEP performance of leading gLMs on the unmatched benchmarks that dominate leaderboards, and about a fifth on matched ones. The confound bites hardest on the smaller single-sequence gLMs, which never exceed it, consistent with Tang et al. (2025), while alignment-based and large models keep signal that survives matching, most of it shared with conservation, the baseline gLM VEP papers should be reporting against. The confound itself is an across-set ascertainment phenomenon, not a local-context artifact.

Limitations. We score zero-shot LLR on SNVs; fine-tuned models and indels are out of scope. The calibration is a shift on a score vector, so it transfers mechanically to a fine-tuned output, but whether it helps there is untested: supervised fine-tuning on ClinVar-style labels should absorb the ascertainment pattern rather than shed it, leaving the coupling in place. The recovery ratio pits a label-fitted classifier against a zero-shot score, making it a *generous* upper bound on compositional overlap, with the label-free 0.31–0.38 as the matching lower one. We cannot separate the surviving signal from conservation (§4.3, §4.6). Composition and function are genuinely correlated; the question is whether a model advertised as learning regulatory grammar reduces to a mutation-rate detector on the benchmarks certifying it. Report composition and conservation baselines alongside gLM VEP, and prefer matched or calibrated scores.

Reproducibility. Code, neutral tables and `calibrate_llr`: https://github.com/rikhinkavuru/composition-confound-vep.

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